



L^AT_EX

Cian Gregg

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1 Introduction

LaTeX, $\text{\LaTeX}$ or even *LAH-tekh*¹ is a Typesetting language used heavily if not exclusively in Maths, Physics and Computer Science. It might look a bit intimidating the first time you see it but its nothing to be scared of. Hopefully after this introduction you'll know how to get yourself started.

2 Structure

When you start a project you'll be faced with something like this

```
\documentclass{article}
\usepackage{graphicx}

\title{\LaTeX}
\author{Cian Gregg}
\date{\today}

\begin{document}

\maketitle

\section{Introduction}
...

\end{document}
```

We have the *preamble*, everything before begin document, this is how we import packages and tweak settings. Over time you'll end up building a collection of packages you like to use, I'll show an example of some of the *core* packages you probably want to be using.

```
%layout
\usepackage{geometry} %%%%% Important, fixes the page settings
\geometry{a4paper, lmargin=2.5cm, rmargin=2.5cm, top=2.5cm, bottom=2.5cm}

%images
\usepackage{graphicx}
\usepackage{float} %%%%% Important, fixed figures spacing with [H]
\usepackage{wrapfig}

%captions
\usepackage{caption}
\captionsetup{figure}{justification=centering}

%subfigures
\usepackage{subcaption}
```

¹which is probably the proper pronunciation but I'm not going to use it

```

%maths fonts
\usepackage{dsfont}
\usepackage{amsfonts}

%maths
\usepackage{amsmath}
\usepackage{amsthm}
\usepackage{amssymb}
\usepackage{fontenc}

%bibilography
\usepackage[sorting=none]{biblatex}
\addbibresource{bib.bib}

% putting indents to zero
\setlength{\parindent}{0cm}

%%% fixes hyperlink colours
\usepackage{hyperref}
\hypersetup{
    colorlinks=true,
    breaklinks=true,
    urlcolor= black,
    linkcolor= black,
    filecolor=black,
    citecolor=black,
    linkbordercolor=black
}

% python display
\usepackage{minted}

```

3 Syntax

Some syntax for writing:

```
\\                                % line break
\newpage                         % new page
\section{Title}                  % section heading
\subsection{Sub Section}        % subsection heading
\subsubsection{Sub Sub Section} % subsubsection heading
\input{file}                    % pull in another .tex file
\maketitle                      % title page
\begin{center}                  % centre environment
    ...
\end{center}
```

3.1 Maths

The main reason we use this typeset language is for its ease of inputting maths equations seamlessly. All you have to do is wrap your maths say, $f(x)=x^2$, in $\$$ for it to render: $f(x) = x^2$.

There's a few conventions for opening maths mode such as $\$$ for inline $\$$ or new line $\\$.

$f(x)=x^2$, $f(x)=x^2$, $[f(x)=x^2]$

Most of the time though you'll open maths environments such as equation or align. Align is one of the most versatile, as the name suggests it lets you align equations with ampersands. These environments typically label each line, its sometimes neater to remove this with a ². A basic example

```
\begin{align*}
    f(x)&=x^2\\
    &=x \cdot x
\end{align*}
```

$$\begin{aligned} f(x) &= x^2 \\ &= x \cdot x \end{aligned}$$

There is genuinely so much you can do in an align environment, some things to get you started:

Special Characters

You can call any Greek or maths symbol in maths mode by using slash its name. $\alpha, \beta, \gamma, \dots$

$\backslash\alpha, \backslash\beta, \backslash\gamma, \backslash\cdots$

A handy site to draw symbols and get their name is detexify.

²adding slash nonumber to each line will do the same

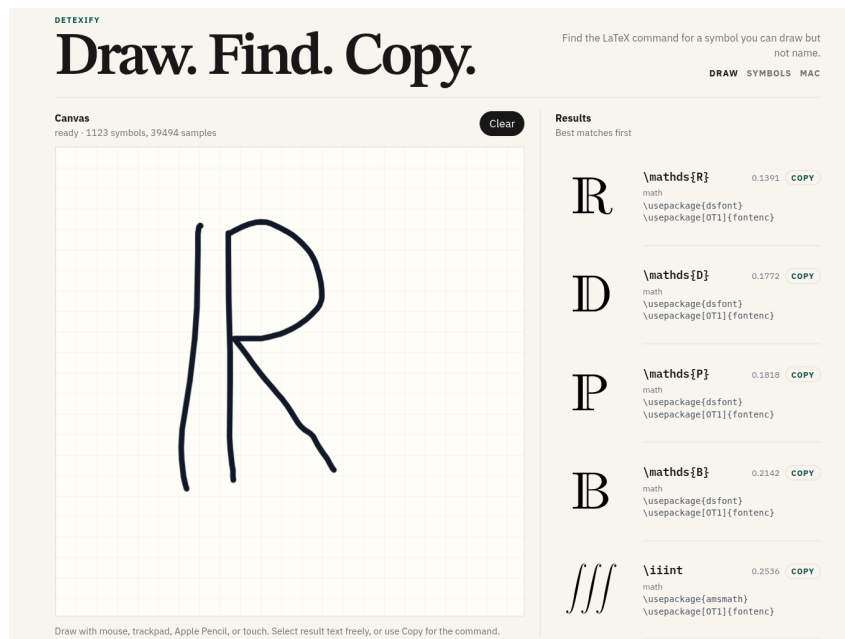


Figure 3.1: Screenshot of detexify

Subscripts and Superscripts

$$y_{\text{Subscripts}}^{\text{Superscripts}}$$

`$y_{\text{\text{Subscripts}}}^{\text{\text{Superscripts}}}$`

Using the Basel problem as an example,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

```
\begin{align*}
\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.
\end{align*}
```

Brackets

With slash left and right you can ensure brackets properly

$$A\left(\frac{a+b}{c}\right) \rightarrow A\left(\frac{a+b}{c}\right)$$

```
\begin{align*}
A\left(\frac{a+b}{c}\right) \rightarrow A\left(\frac{a+b}{c}\right)
\end{align*}
```

Matrices

Two useful options are *pmatrix* (Parentheses) and *bmatrix* (Brackets)

$$\begin{pmatrix} ye^x \\ \cos(xy) \end{pmatrix} \qquad \begin{bmatrix} ye^x \\ \cos(xy) \end{bmatrix}$$

```

\begin{align*}
&\begin{pmatrix}
ye^x \\
\cos(xy)
\end{pmatrix} && \\
&\begin{bmatrix}
ye^x \\
\cos(xy)
\end{bmatrix} \\
\end{align*}

```

So much more

There's so so much more we can do with LaTeX: defining your own commands, tikz designs, but for my own laziness I'll link [overleaf's handy guide](#) and stop listing stuff. Since I have it handy, here's some analysis assignment from last year as an example:

Applications of the inverse function theorem

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the map

$$f(x, y) = \begin{pmatrix} ye^x \\ \cos(xy) \end{pmatrix}$$

Determine the derivative $(Df)_{(x,y)}$ at all points at which f is differentiable.

Proof. Let $\vec{p} \in \mathbb{R}^2$

$$\begin{aligned}
((Jf)_{\vec{p}})_{ij} &= \partial_j f_i(\vec{p}) \\
(Jf)_{\vec{p}} &= \begin{pmatrix} \partial_x f_1 & \partial_y f_1 \\ \partial_x f_2 & \partial_y f_2 \end{pmatrix} = \begin{pmatrix} \partial_x ye^x & \partial_y ye^x \\ \partial_x \cos(xy) & \partial_y \cos(xy) \end{pmatrix} \\
&= \begin{pmatrix} ye^x & e^x \\ -y \sin(xy) & -x \sin(xy) \end{pmatrix}
\end{aligned}$$

Since the components of the Jacobian are all continuous for any arbitrary $\vec{p} \in \mathbb{R}^2$, the Jacobian is continuous on $\mathbb{R}^2$. That implies that f is differentiable everywhere on $\mathbb{R}^2$ and that the derivative is defined $(Jf)_{\vec{p}} = (Df)_{\vec{p}}$

$$(Df)_{(x,y)} = \begin{pmatrix} ye^x & e^x \\ -y \sin(xy) & -x \sin(xy) \end{pmatrix}$$

□

3.2 Figures

This is probably the most *fiddly* part of LaTeX but with *float* its grand. Also overleaf has a handy insert figure button which is handy for getting the formatting correct. An example of some MCMC Ising model work,

```

\begin{figure}[H]
  \centering
  \includegraphics[width=0.7\linewidth]{images/observables... .pdf}
  \caption{Wolf Alogrithm running on a  $100 \times 100$  Ising Model....}
  \label{fig:wolff}
\end{figure}

```

This will produce Figure 3.2, which we can reference with slash ref{fig:wolff}. A few things to note is the use of [H] this enforced that the figure is placed **exactly** where the begin figure is placed, otherwise Latex will fit it in the next available place which is usually miles away. Another thing is the use of *pdfs*, they're so much higher quality and make reports look so much better.

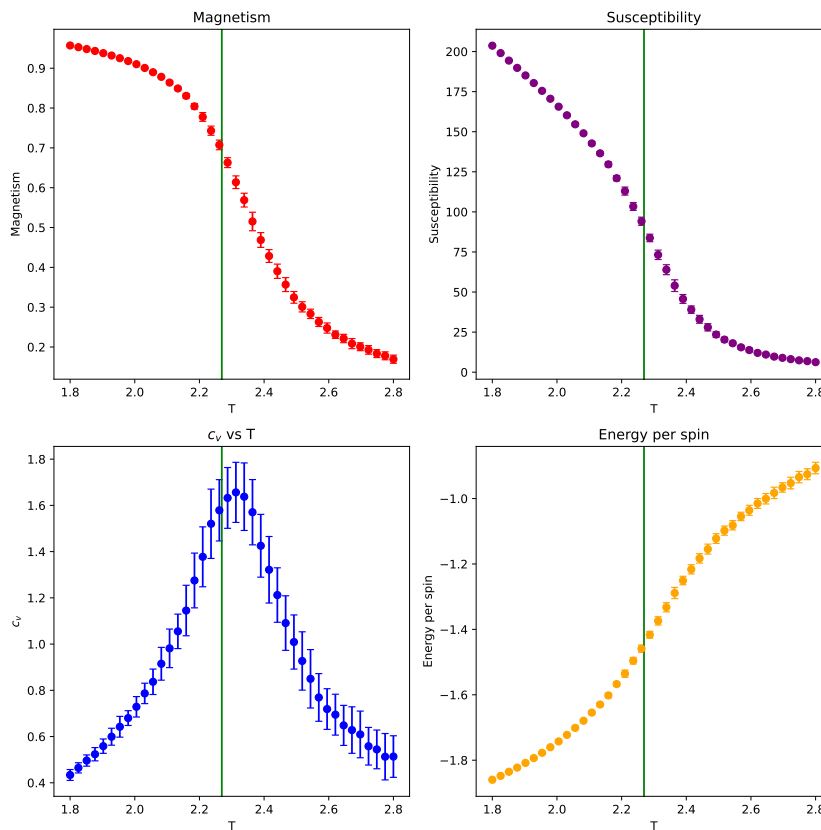


Figure 3.2: Wolf Alogrithm running on a 100×100 Ising Model to calculate observables

There's more fancy stuff like sub captions,

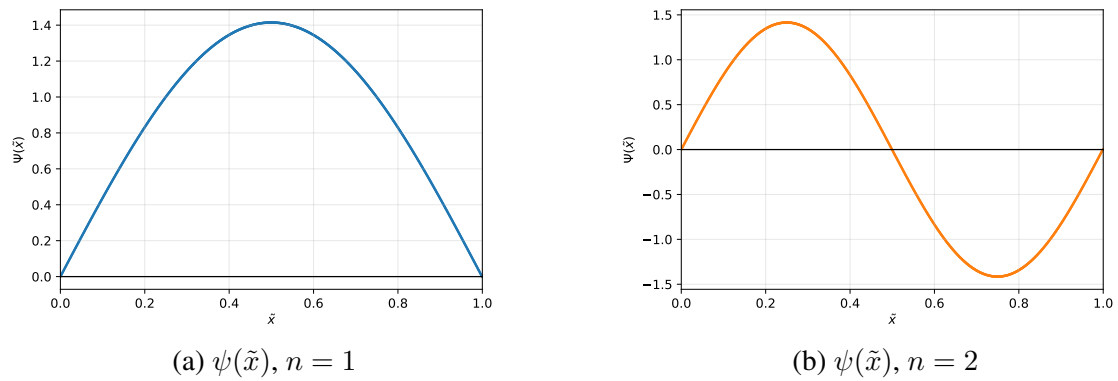


Figure 3.3: Square well eigenstates

3.3 Python

Especially for the likes of the computational labs, printing out code is a nice touch. There's tonnes of packages for this personally my favourite is *minted*.

A quick example,

```
...
T=Tstart
while T > Tend:
    for _ in range(sweeps_per_T):
        E = sweep_once(s, N, J, h, T, E)
        energies.append(E)
    T *= alpha
...
```

This environment has a many different languages and different settings to adjust, I've left it pretty plain here.

```
\begin{minted}[fontsize=\normalsize]{python}
...
\end{minted}
```

4 Referencing

There is 100% more sophisticated ways of going about this, however this has been enough for lab reports over the last few years. We're using *biblatex*. In our preamble we have,

```
\usepackage[sorting=none]{biblatex}
\addbibresource{bib.bib}
```

This takes in a bibliography .bib file for our project. The easiest way to build this is with Google Scholar, just type in the name book or article and then select BibTex, see Figure 4.1.

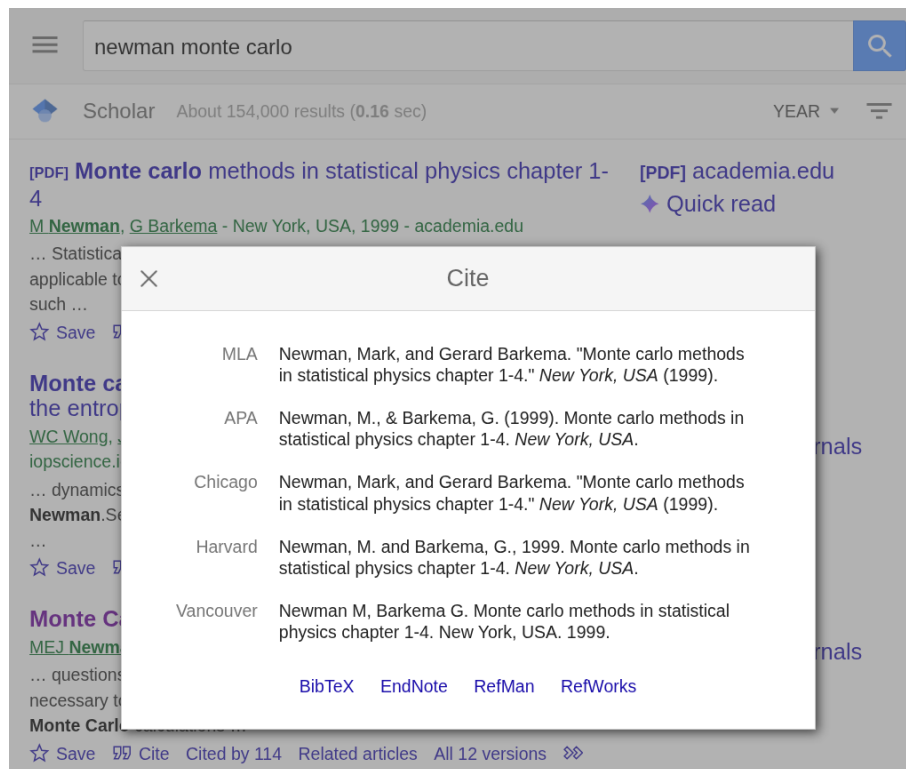


Figure 4.1: Using Google Scholar

Using some textbooks I’ve got handy as an example

```
@article{newman1999,
  title={Monte carlo methods in statistical physics},
  author={Newman, and Barkema},
  journal={New York, USA},
  year={1999}
}
@book{landau2021guide,
  title={A guide to Monte Carlo simulations in statistical physics},
  author={Landau, David and Binder, Kurt},
  year={2021},
  publisher={Cambridge university press}
}
```

Now with these in our .bib file we can cite Newman [1] and Landau [2] with

```
\cite{newman1999} and \cite{landau2021guide}
```

References

- [1] Newman and Barkema. “Monte carlo methods in statistical physics”. In: *New York, USA* (1999).
- [2] David Landau and Kurt Binder. *A guide to Monte Carlo simulations in statistical physics*. Cambridge university press, 2021.

5 Bonus

Links

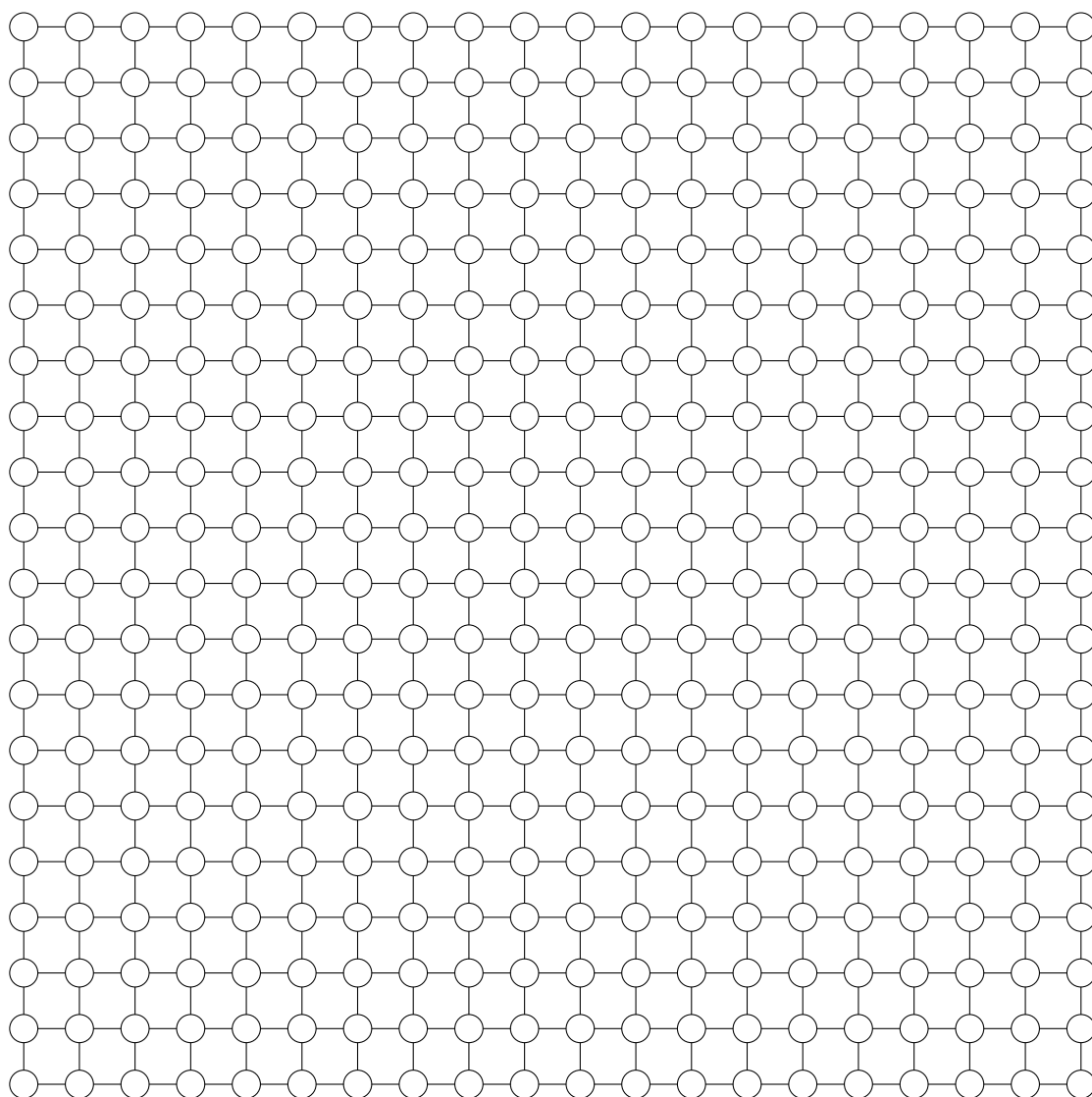
[Tables Generator](#)

[LaTeX to Image](#)

Extras

TikZ package

Here's a 20×20 square lattice I pointlessly made.



Good Luck!

Good luck with your assignments and lab reports, here's [Overleaf's Documentation](#) if you've got any more questions.